

Lab Meeting

2024/7/9

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 - Uncertainty estimation for trajectory prediction
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 - Apply KAIST EE
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Previous Notes

Previous Notes

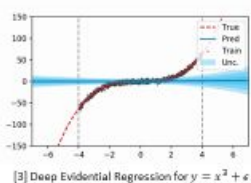
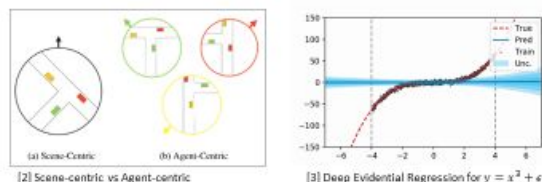
- Uncertainty Based Control
 - Trajectory Prediction with **evidential deep learning**
 - How can we trust estimated value α (epistemic uncertainty)?
 - Choose another reliable method!

Current Works

• Uncertainty Based Control

• Future Plan

- 1) Train Agent-Centric Trajectory Prediction Model with **evidential deep learning**
- 2) Replace Trajectron++ to 1)
- 3) Add the **model uncertainty** as a new constraints for the agent control and planning

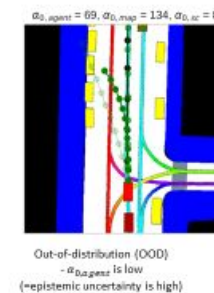


[2] Aydemir, G., Akan, A. K., & Güney, F. (2023). Adapt: Efficient multi-agent trajectory prediction with adaptation. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 8295-8305).
[3] Amini, A., Schwarting, W., Soleimany, A., & Rus, D. (2020). Deep evidential regression. Advances in neural information processing systems, 33, 14927-14937.

Current Works

Interpretable Self-Aware Prediction

- Trained at slow speed agents (ID).
- Fast moving objects are considered as OOD.

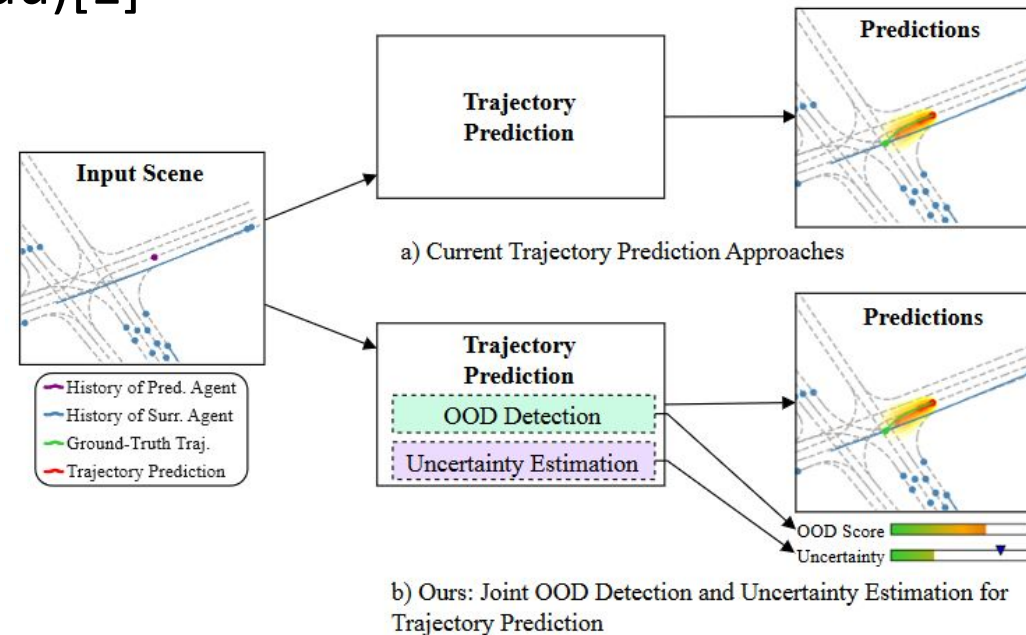


[4] Itkina, M., & Kochenderfer, M. (2023, March). Interpretable self-aware neural networks for robust trajectory prediction. In Conference on Robot Learning (pp. 606-617). PMLR.

Current Works

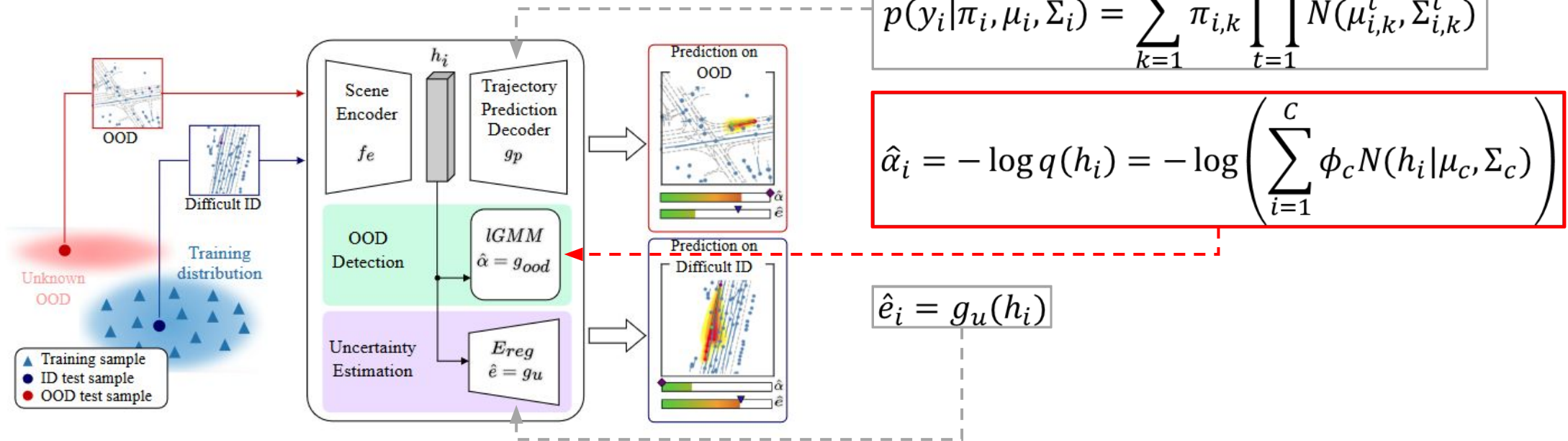
Current Works

- Trajectory prediction model
 - Joint Out-of-Distribution Detection and Uncertainty Estimation for Trajectory Prediction (Joodu)[1]



Current Works

- Trajectory prediction model
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Cu

Assume that probability distribution of the latent features follows
a mixture of multivariate Gaussian distributions [2]

• Tra

Gaussian mixture model $q(h_i) = \sum_{c=1}^C \phi_c N(h_i | \mu_c, \Sigma_c)$

OOD detector (latent GMM; IGMM) outputs the OOD score

$$\hat{\alpha}_i = -\log q(h_i) = -\log\left(\sum_{c=1}^C \phi_c N(h_i | \mu_c, \Sigma_c)\right)$$

as the negative log-likelihood under the Gaussian mixture distribution $q(h_i)$

- ID scenario has a low score
- OOD scenario has a high score

Unknown
OOD

▲ Training
● ID test s
● OOD tes

[1] Wiederer, J., Schmidt, J., Kressel, U., Dietmayer, K., & Belagiannis, V. (2023, October). Joint Out-of-Distribution Detection and Uncertainty Estimation for Trajectory Prediction. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 5487-5494). IEEE.

[2] Lee, K., Lee, K., Lee, H., & Shin, J. (2018). A simple unified framework for detecting out-of-distribution samples and adversarial attacks. Advances in neural information processing systems, 31.

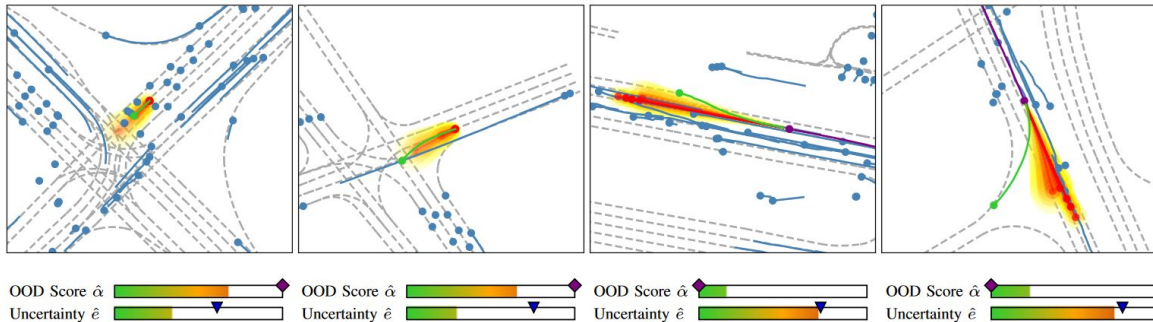
Current Works

- Trajectory prediction model
 - Joint Out-of-Distribution Detection and Uncertainty Estimation for Trajectory Prediction [1]

Trajectory Prediction Results (SHIFTS eval set)

Model	Scene rast.	Input vect.	Ensemble	wADE ↓			minADE ↓			wFDE ↓			minFDE ↓			NLL ↓		
				ID	OOD	Full	ID	OOD	Full	ID	OOD	Full	ID	OOD	Full	ID	OOD	Full
BC (K=1) [1]	•			1.104	1.407	1.164	0.829	1.084	0.880	2.394	3.197	2.555	1.733	2.420	1.870	106.15	144.77	113.88
BC (K=5) [1]	•		•	1.028	1.299	1.082	0.777	1.014	0.824	2.238	2.957	2.382	1.636	2.278	1.765	103.70	140.95	111.15
DIM (K=1) [1]	•			1.551	1.883	1.618	0.759	0.942	0.796	3.536	4.376	3.704	1.511	1.983	1.605	96.45	121.95	101.55
DIM (K=5) [1]	•		•	1.424	1.754	1.490	<u>0.728</u>	<u>0.918</u>	0.766	3.256	4.093	3.424	<u>1.493</u>	2.000	1.595	97.14	124.80	102.68
VNT [10]		•		-	-	1.326	-	-	0.495	-	-	3.158	-	-	0.936	-	-	61.55
ViT [11]	•			-	-	1.850	-	-	0.526	-	-	4.433	-	-	1.016	-	-	<u>61.63</u>
HiVT*		•		<u>1.031</u>	1.234	1.072	0.394	0.473	0.410	<u>2.367</u>	2.933	<u>2.480</u>	0.748	0.968	0.792	-20.42	-16.44	-19.62

Qualitative results

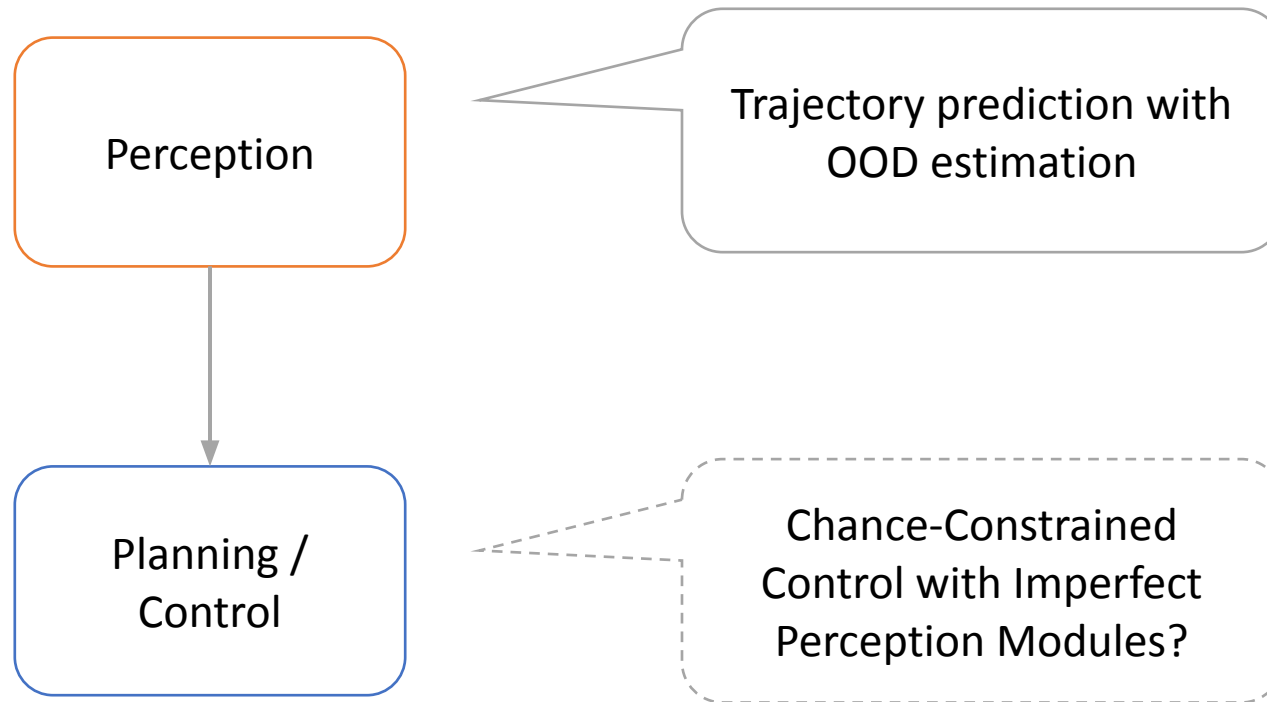


OOD Detection / Uncertainty Estimation Results

Prediction Model	$\hat{\alpha}$	$\hat{\epsilon}$	AUROC (%) ↑	Uncertainty Estimation $\hat{\epsilon}$		
				ID	OoD	Full
BC (K=1) [1]	MA		52.8	-	-	0.293
BC (K=5) [1]	MA		52.1	-	-	0.258
DIM (K=1) [1]	MA		51.8	-	-	0.458
DIM (K=5) [1]	MA		50.9	-	-	0.411
VNT [10]	SNGP		54.0	-	-	0.327
ViT [11]	E_{reg}		53.1	-	-	0.455
HiVT*	NLL		50.3	<u>0.232</u>	0.293	0.244
HiVT*	MA		54.6	0.421	0.525	0.440
HiVT*	SNGP		52.6	0.253	0.316	0.265
HiVT*	E_{reg}		52.3	0.222	0.272	0.232
HiVT*	IGMM		56.8	0.436	0.462	0.440
HiVT*	IGMM	E_{reg}	56.8	0.222	0.272	0.232

Current Works

- Related works for research positioning



Current Works

- Apply KAIST EE



지원서작성

마이페이지

제출서류관리

합격자발표

Home > 마이페이지



My Page

KAIST, 세계의 중심에서! 실패 없는 노력과 끝없는 도전이 계속되는 KAIST는 열정입니다.

• 지원자 정보

전형구분	2025 학년도 봄학기 1차	지원과정	석사
지원자명	함형찬	수험번호	110959
지원분야	전기및전자공학부	학생구분	• 1선택 : 국비 또는 KAIST

비밀번호 변경

입학원서 출력

수험표 출력

발송용봉투 출력

입학원서확인

입학원서수정

※ 수험표 등 면접에 필요한 자료는 2024.08.09 금요일까지 출력하시기 바랍니다. 평가결과등록기간 2024.08.10 토요일 부터 최종합격자 발표 이전까지는 로그인 이 불가능하므로 필요한 서류는 미리미리 준비하여 주시기 바랍니다.

※ Internet Explorer 에서 출력이 원활하게 안 될 경우 크롬(Chrome) 브라우저를 이용하여 출력하여 주시기 바랍니다.

• keyword /자기소개서

Keyword 작성여부	작성완료	등록 (수정)
자기소개서 작성여부	작성완료	등록 (수정)

• 전형료 납부내역

전형료 납부	100,000 원
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Current Works

- KAIST individual study plan (TBD)

최준명

(W1) Kick-off meeting

(W2) rlkit framework [3] 환경 구현

(W3) baseline (DQN)으로 cart-pole 환경 학습

(W4) CQL [5] 구현

(W5) ...

(W6) ...

(W7) ...

(W8) ...

최종 기대

- 각종 강화학습 프레임워크 구현
- 강화학습 모델 재연
- 선정 논문 재연
- ...

한의석

(W1) Kick-off meeting

(W2) rllab framework [4] 환경 구현

(W3) baseline (TRPO)으로 cart-pole 환경 학습

(W4) CPO [6] 구현

(W5) ...

(W6) ...

(W7) ...

(W8) ...

더 해볼 것들

- Cart-pole based on MPC?
- Ablation study?
- Apply to various environments?
- ...

[3] <https://github.com/rail-berkeley/rlkit>

[4] <https://github.com/rll/rllab>

[5] Kumar, A., Zhou, A., Tucker, G., & Levine, S. (2020). Conservative q-learning for offline reinforcement learning. Advances in Neural Information Processing Systems, 33, 1179-1191.

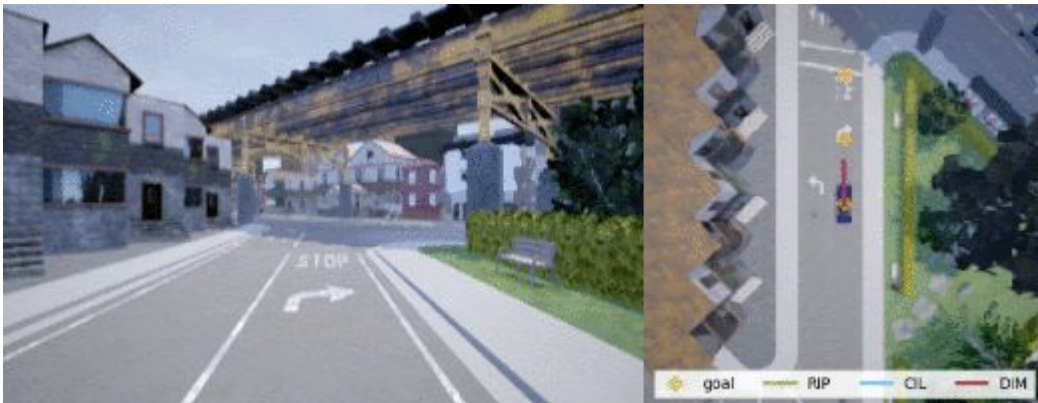
[6] Achiam, J., Held, D., Tamar, A., & Abbeel, P. (2017, July). Constrained policy optimization. In International conference on machine learning (pp. 22-31). PMLR.

Future Works

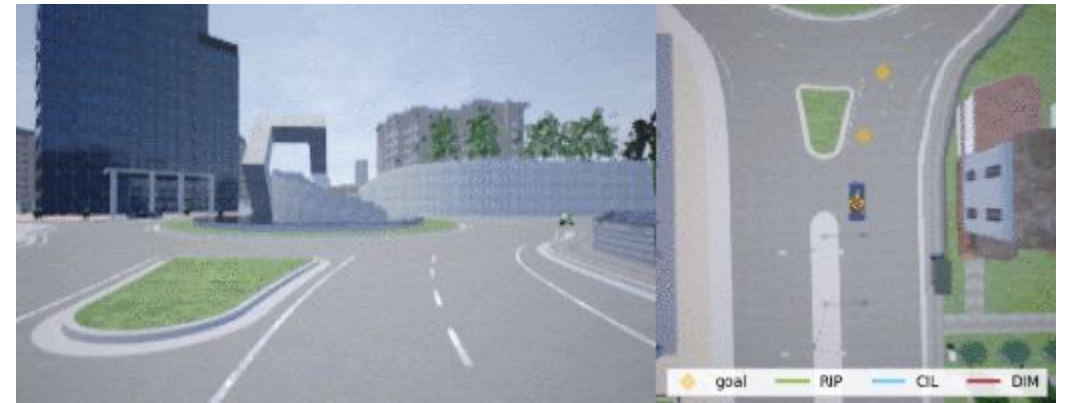
Future Works

- More Survey...
 - Topic: learning-based control, OOD autonomous driving,

Robust imitative planning (RIP): epistemic uncertainty-aware planning method (w/ imitation learning)



AbnormalTurns0-v0



Roundabouts2-v0